

The procedure for obtaining the product of maxterms directly from the truth table is as follows. Form a maxterm for each combination of the variables which produces a 0 in the function, and then form the AND of all those maxterms.

Boolean functions expressed as a sum of minterms or product of maxterms are said to be in canonical form.

2.5.2 Sum of Minterms

Example 2.5: Express the Boolean function $F = A + B'C$ in a sum of minterms.

$$A + B'C = A \cdot 1 + B'C \quad (\text{Postulate 2(b)})$$

$$= A \cdot (B + B') + B'C \quad (\text{Postulate 5(a)})$$

$$= (AB + AB') + B'C \quad (\text{Postulate 4(a)})$$

$$= (AB \cdot 1 + AB' \cdot 1) + (B'C) \cdot 1 \quad (\text{Postulate 2(b)})$$

$$= (AB \cdot (C + C') + AB' \cdot (C + C')) + B'C \cdot (A + A') \quad (\text{Postulate 5(a)})$$

$$= (ABC + ABC') + (AB'C + AB'C') + B'C \cdot A + B'C \cdot A' \quad (\text{Postulate 4(a)})$$

$$= (ABC + ABC') + (AB'C + AB'C') + (AB'C + AB'C') \quad (\text{Postulate 3(b)})$$

$$= (ABC + ABC') + (AB'C + AB'C') + (AB'C + AB'C') \quad (\text{Theorem 4(a)})$$

$$= (ABC + ABC') + ((AB'C + AB'C') + AB'C + AB'C') \quad (\text{Theorem 4(a)})$$

$$= (ABC + ABC') + ((AB'C + AB'C') + A'B'C) \quad (\text{Postulate 3(a)})$$

$$= (ABC + ABC') + ((AB'C + AB'C') + A'B'C) \quad (\text{Theorem 4(a)})$$

$$= (ABC + ABC') + ((AB'C + AB'C') + A'B'C) \quad (\text{Theorem 1(a)})$$

$$= (ABC + ABC') + ((AB'C + AB'C') + A'B'C) \quad (\text{Theorem 4(a)})$$

$$= ((ABC + ABC') + AB'C) + (AB'C' + A'B'C') \quad (\text{Theorem 4(a)})$$

$$= ((ABC + ABC') + AB'C) + (AB'C' + A'B'C') \quad (\text{Theorem 4(a)})$$

$$= ((ABC + ABC') + AB'C) + (AB'C' + A'B'C') \quad (\text{Theorem 4(a)})$$

$$= ABC + ABC' + AB'C + AB'C' + A'B'C'$$

$$= ABC + ABC' + AB'C + AB'C' + A'B'C'$$

Rearranging the minterms in ascending order, we finally obtain:

$$F = A'B'C' + AB'C' + AB'C + ABC' + ABC$$

$$= m_1 + m_4 + m_5 + m_6 + m_7$$

$$F(A, B, C) = \sum(1, 4, 5, 6, 7)$$